EE561 Advanced Static Power Conversion

Multilevel Converters: Building the Levels — Quiz Answer Key

Sefer İdacı · for the presenter and the instructor · not for distribution to students

Question 1 · Levels — 25 pts

(a) Step = voltage on each switch = $V_{dc}/(N-1)$; switches per leg = $2(N-1)$.

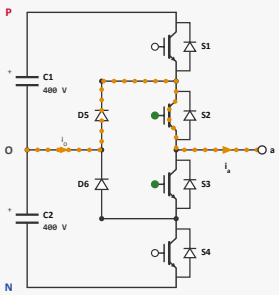
Levels N	Step = voltage blocked by each switch	Switches in one leg
2	**800 V**	**2**
3	**400 V**	**4**
5	**200 V**	**8**

(b) Any two of each (slides 5 and 6).
Advantages: lower voltage on each device (lower-voltage, faster parts); smaller voltage steps (less dv/dt and EMI); smaller current ripple (smaller filter); lower switching loss.
Disadvantages: more switches and gate drivers; more devices in series (more conduction loss); capacitor voltages that must be kept balanced; more switching states (harder control).

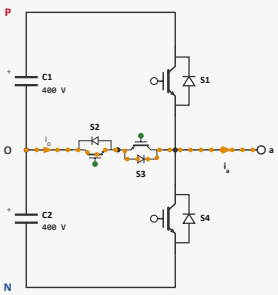
Marking: (a) 2.5 pts per cell; (b) 2.5 pts per correct item, at most two of each kind.

Question 2 · Reading the circuits — 25 pts

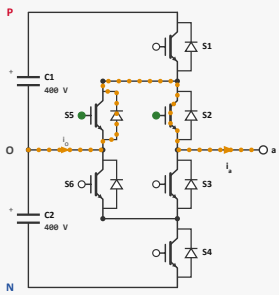
(a) The current paths ($i_a > 0$, from the midpoint O to the output):



A: NPC · O → D5 → S2 → a



B: T-type · O → S2 → diode of S3 → a



C: ANPC · O → diode of S5 → S2 → a

(b) They are **redundant states**: different switching states with the same output voltage. In D (state 10: S1 and S2′ on) the load current flows into the + terminal of C_{fc} and **charges** it; in E (state 01: S2 and S1′ on) it flows the other way and **discharges** it. Choosing between them keeps V_{fc} at $V_{dc}/2$.

Marking: (a) path 3, name 2 per leg; (b) “redundant” 4, charge / discharge 6.